

Different Questions (Not in Our Question Bank)

1. 1. Total body Na⁺ content — Na⁺ is the main osmotically active ECF solute; ECF volume tracks total body sodium.
2. 2. Homeostasis — maintenance of a stable internal environment despite external change, via feedback control.
3. 3. Nucleus tractus solitarius (NTS) — receives vagal afferents from gastric stretch receptors.
4. 4. The cell — smallest structural/functional unit of life.
5. 5. Mitochondria — site of oxidative phosphorylation/ATP synthesis.
6. 6. Gap junction — connexin (tight junction— claudin/occludin; desmosome— cadherin/desmoplakin).
7. 7. Sodium (Na⁺) — principal extracellular cation.
8. 8. Intracellular fluid (ICF) — ~2/3 of total body water.
9. 9. ICF is K⁺/protein-rich; ECF is Na⁺/Cl⁻-rich — maintained by Na⁺/K⁺-ATPase.
10. 10. Na⁺/K⁺-ATPase — primary active transporter (3 Na⁺ out, 2 K⁺ in per ATP).
11. 11. Simple diffusion — not carrier-mediated, so no saturation kinetics.
12. 12. Na⁺/K⁺-ATPase pumps 3 Na⁺ out, 2 K⁺ in per ATP — sets up the ionic gradients/RMP.
13. 13. Rapid Na⁺ influx via voltage-gated Na⁺ channels — drives depolarization.
14. 14. K⁺ efflux via voltage-gated (delayed rectifier) K⁺ channels — causes repolarization.
15. 15. Na⁺ channel inactivation (± persistent K⁺ conductance) — underlies the refractory period.
16. 16. H-band and I-band shorten (A-band length unchanged) during contraction.
17. 17. The Na⁺-channel-inactivated phase (depolarization spike through most of repolarization) = absolute refractory period.
18. 18. Failure of voltage-gated K⁺ (delayed rectifier) channels to open.
19. 19. Center of the A-band, where thin filaments overlap during full contraction.
20. 20. Binding of a new ATP molecule to the myosin head.

21. 21. Troponin — present in skeletal/cardiac muscle, absent in smooth muscle (which uses calmodulin/caldesmon).
22. 22. Pi release from myosin causes the head to pivot, pulling actin toward the sarcomere center (power stroke).
23. 23. T-tubule depolarization is sensed by DHP receptors, mechanically coupled to RyR1 to release SR Ca^{2+} (no need for extracellular Ca^{2+} entry).
24. 24. SERCA pumps Ca^{2+} back into the SR, lowering cytosolic Ca^{2+} , ending contraction.
25. 25. Troponin C — binds Ca^{2+} to initiate cross-bridging.
26. 26. Cardiac muscle needs extracellular Ca^{2+} influx (CICR) to trigger SR release; skeletal muscle doesn't.
27. 27. ACh binds nicotinic receptors on the motor end plate, generating an end-plate potential.
28. 28. Gamma (γ) motor neurons — innervate intrafusal fibers.
29. 29. GABA (also glycine) — main inhibitory CNS neurotransmitter.
30. 30. Directly proportional — thicker, myelinated fibers conduct faster.
31. 31. The neuron — functional unit of the nervous system.
32. 32. Glutamate — main excitatory CNS neurotransmitter.
33. 33. Neural (autonomic) and endocrine (hypophyseal) control, integrating feedback signals.
34. 34. Cerebellum — coordinates voluntary movement and posture.
35. 35. Initiating/planning movement and suppressing unwanted movement — basal ganglia.
36. 36. Emotion, memory, and motivation — limbic system.
37. 37. Chemical synapses — transmission is one-way (pre- to postsynaptic).
38. 38. Acetylcholine — from all preganglionic autonomic fibers (nicotinic receptors).
39. 39. Norepinephrine — from postganglionic sympathetic fibers (except sweat glands = ACh).
40. 40. $A\alpha > A\beta > A\gamma > A\delta > B > C$ — descending conduction velocity.
41. 41. Caudate nucleus (also putamen, globus pallidus, substantia nigra, subthalamic nucleus).

42. 42. Sympathetic fibers diverge widely and also trigger adrenal medullary catecholamine release, which is cleared slowly from blood.
43. 43. β_2 -adrenergic receptors — bronchial smooth muscle relaxation.
44. 44. Adrenal medulla, sweat glands, most blood vessels, piloerectors — sympathetic-only innervation.
45. 45. Contracts detrusor, relaxes internal sphincter — promotes micturition.
46. 46. Miosis (pupil constriction) is NOT sympathetic — it's parasympathetic; sympathetic causes mydriasis.
47. 47. Passive electrotonic (decremental) conduction with summation at the soma.
48. 48. Receptor → afferent neuron → CNS integration center → efferent neuron → effector.
49. 49. Monosynaptic reflex — stretch/myotatic reflex.
50. 50. Muscle spindle (Ia afferents) — detects length and rate of stretch.
51. 51. Golgi tendon organ — senses muscle tension, provides protective autogenic inhibition via Ib afferents.
52. 52. Ca^{2+} -calmodulin activates MLCK, which phosphorylates myosin light chains to permit cross-bridging.
53. 53. Exaggerated extensor (decerebrate) rigidity — loss of inhibitory descending input.
54. 54. Ib afferents from Golgi tendon organs — mediate the inverse stretch reflex.
55. 55. Submandibular and otic ganglia — parasympathetic supply to salivary glands.
56. 56. Iris — colored part of the eye.
57. 57. Pacinian corpuscle — rapidly-adapting (phasic) receptor.
58. 58. Olfactory epithelium (nose) — detects odors.
59. 59. Eustachian tube — drains middle/inner-ear-related fluid to the nasopharynx.
60. 60. Labeled-line principle — modality is coded by which dedicated pathway/receptor fires, not the signal itself.
61. 61. Receptor/channel desensitization — declining generator potential despite constant stimulus.
62. 62. Increased firing frequency (rate coding) and recruitment, not AP amplitude.
63. 63. Slowly-adapting (tonic) receptors — e.g., nociceptors, muscle spindles, baroreceptors.

64. 64. Light passes through ganglion and bipolar cell layers first, reaching photoreceptors last (inverted retina).
65. 65. Superior temporal gyrus (Heschl's gyrus) — auditory cortex.
66. 66. K^+ influx into hair cells (endolymph is K^+ -rich) — initiates depolarization.
67. 67. Left homonymous hemianopia — right optic tract lesion.
68. 68. Place theory / tonotopic organization of the basilar membrane.
69. 69. Difficulty localizing sound / processing tonal- musical information (hearing loss is usually mild/bilateral cortical input).
70. 70. Ciliary muscle relaxes, lens flattens — reduced accommodation for distance viewing.
71. 71. Reduced rhodopsin (bleaching) and reduced cGMP-channel sensitivity — light adaptation.
72. 72. Contralateral superior homonymous quadrantanopia ("pie in the sky") — Meyer's loop lesion.
73. 73. Hemoglobin (in RBCs) — main O_2 carrier.
74. 74. Systemic (non-renal, non-testicular) capillary beds lack a countercurrent mechanism (unlike vasa recta/testis).
75. 75. Fibrinogen (and other clotting factors) — present in plasma, absent in serum.
76. 76. Anemia / CO poisoning / decreased Hb — reduce O_2 -binding capacity.
77. 77. HbF ($\alpha_2\gamma_2$) — predominant fetal hemoglobin.
78. 78. HbF — highest O_2 affinity (binds 2,3-BPG poorly).
79. 79. Decreased 2,3-BPG ($\uparrow$ Hb- O_2 affinity) and increased extracellular K^+ — changes in stored blood.
80. 80. Electrolytes, albumin, glucose — present in both serum and plasma (fibrinogen/clotting factors are plasma-only).
81. 81. A child with type O is genetically possible from A(het)×B(het) parents — not evidence against parentage.
82. 82. HbF — lowest 2,3-BPG affinity.
83. 83. $\downarrow$ Temperature, $\downarrow$ 2,3-BPG, $\downarrow$ CO $_2$ /alkalosis, HbF/CO presence — shift curve left.
84. 84. Higher CO $_2$ /H $^+$ in venous blood (Bohr effect) lowers Hb- O_2 affinity.
85. 85. Local vasoconstriction (vasospasm) — initial response to vessel injury.

86. 86. von Willebrand factor bridging platelet GPIb to exposed collagen.
87. 87. GPIIb/IIIa receptor (binds fibrinogen) — platelet- platelet aggregation.
88. 88. Factor VII has the shortest half-life of the vitamin K–dependent factors, so PT (extrinsic) prolongs first.
89. 89. Platelet activation — shape change, granule release, and aggregation (positive feedback).
90. 90. Protein C/Protein S (inactivates Va, VIIIa) — also antithrombin III and TFPI.
91. 91. Biconcave, anucleate discs — erythrocyte morphology.
92. 92. Carbohydrate (glycoprotein/glycolipid) antigens on the RBC surface, set by glycosyltransferase genes.
93. 93. Lacks A, B, and Rh(D) antigens — won't trigger ABO/Rh reactions in any recipient.
94. 94. Lewis antigens (plasma-derived, adsorbed onto RBCs) can show altered expression with abnormal plasma lipoproteins in liver disease.
95. 95. Anti-A/anti-B are pre-formed IgM antibodies present without prior exposure, causing immediate hemolysis; Rh needs prior sensitization.
96. 96. Homozygous loss-of-function mutation in the H (FUT1) gene — no H antigen, so A/B can't be expressed.
97. 97. AV node — acts as the heart's "gate," delaying/filtering impulses from atria to ventricles.
98. 98. Arterioles — greatest drop in systemic arterial BP.
99. 99. Ca^{2+} influx (L-type channels) — maintains the cardiac plateau (Phase 2).
100. 100. Isovolumetric contraction (also true of isovolumetric relaxation) — both valve sets closed, volume constant.
101. 101. Stimulates gastric HCl/pepsinogen secretion and gastrin release — vagal cephalic phase.
102. 102. Loss of vagally-mediated receptive relaxation — impairs liquid emptying after vagotomy.
103. 103. Hypotonic, low-volume, mucous-rich secretion at rest.
104. 104. Peristaltic ("myenteric") reflex — contraction proximal, relaxation distal to a bolus.
105. 105. Ductal cells reabsorb Na^+/Cl^- (duct impermeable to water) faster than they secrete $\text{K}^+/\text{HCO}_3^-$ — dilutes saliva.

106. 106. Cholecystokinin (CCK) — inhibits gastric emptying when fat enters duodenum.
107. 107. HCl (H⁺) secretion is impaired — carbonic anhydrase supplies H⁺ in parietal cells.
108. 108. Secretin release — stimulates pancreatic HCO₃⁻, inhibits gastric acid/motility.
109. 109. Gastric parietal cells — source of intrinsic factor, required for B12 absorption in ileum.
110. 110. Cholecystokinin (CCK) — stimulated by duodenal amino acids/peptides.
111. 111. Na⁺-glucose cotransport (SGLT1) apical, GLUT2 basolateral.
112. 112. Highly acidic (pH ~1–2); contains HCl, pepsinogen, intrinsic factor, mucus.
113. 113. Activates trypsinogen → trypsin, which then activates other pancreatic proenzymes.
114. 114. Triglyceride hydrolysis is impaired (lipase needed before micelle formation/absorption).
115. 115. GLUT5 — apical fructose transporter (facilitated diffusion).
116. 116. Emulsify fat and form micelles to solubilize lipids/fat-soluble vitamins for absorption.
117. 117. Trypsinogen (also chymotrypsinogen, procarboxypeptidase, prolipase) — secreted inactive.
118. 118. Secretin — stimulates pancreatic ductal HCO₃⁻ secretion.
119. 119. Detoxification/drug metabolism and plasma protein synthesis — liver functions unrelated to digestion.
120. 120. Conjugated hyperbilirubinemia, pale stools, dark urine, steatorrhea — common bile duct obstruction.
121. 121. Carbohydrates (RQ = 1.0) — highest respiratory quotient.
122. 122. Basal metabolic processes (mainly Na⁺/K⁺-ATPase activity across organs) — major resting energy expenditure.
123. 123. Renal corpuscles/glomeruli — located in the cortex, not medulla.
124. 124. Vasodilation/natriuresis — NOT effects of Angiotensin II (it vasoconstricts, releases aldosterone, promotes thirst/ADH).
125. 125. ~125 mL/min — normal GFR.
126. 126. Distal tubule and collecting duct — site of urine acidification (H⁺ secretion by intercalated cells).

127. 127. Pancreatic (secretin-stimulated) bicarbonate — neutralizes duodenal HCl.
128. 128. Thick ascending limb of the loop of Henle — impermeable to water under all conditions.
129. 129. Inulin — freely filtered, not reabsorbed/secreted; its clearance = GFR, same behavior as fully- blocked glucose.
130. 130. ~600–700 mL/min — effective renal plasma flow.
131. 131. Efferent arteriolar constriction — most effectively raises GFR.
132. 132. Increased renin secretion — direct JGA response to low pressure/low.
133. 133. Reduced GFR decreases clearance of urea/creatinine, causing accumulation.
134. 134. Maintained relatively constant (80–180 mmHg MAP) via myogenic + tubuloglomerular feedback, independent of nerves/hormones.
135. 135. Regulates blood pressure/volume via Na⁺ and water retention plus vasoconstriction.
136. 136. ↓Renal perfusion pressure, ↓ NaCl at macula densa, ↑sympathetic (β1) activity — stimulate renin.
137. 137. Aldosterone — increases urinary K⁺ excretion.
138. 138. Reduced GFR / mild renal impairment.
139. 139. Volume of plasma completely cleared of a substance per unit time.
140. 140. Kidneys reabsorb filtered HCO₃⁻ and excrete titratable acid/NH₄⁺, acting over hours-to-days.
141. 141. The glomerulus — damage causes proteinuria + hematuria.
142. 142. Urea, creatinine, some ammonia present; no glucose/protein/blood/ketones — normal urine.
143. 143. Hypertension, hypokalemia, metabolic alkalosis, low renin — Conn's syndrome.
144. 144. GFR maintained/slightly ↑ , RPF ↓ — filtration fraction increases.
145. 145. Effective renal plasma flow (RPF) — represented by such a substance's clearance (e.g., PAH).
146. 146. Baroreceptor (↓ stretch), macula densa (↓ NaCl), and β1-sympathetic signals trigger JG cell renin release.
147. 147. Non-specific, immediate defenses present from birth (barriers, phagocytes, complement, NK cells) — innate immunity.

148. 148. Antigen specificity and immunological memory — distinguishes adaptive from innate.
149. 149. T lymphocytes — cell-mediated immunity.
150. 150. ~60–70% of MVV — dyspnea appears when ventilatory reserve (dyspneic index) falls to about this level.
151. 151. ATP-PCr (phosphagen) system — 100m dash energetics.
152. 152. Antigen-specific lymphocytes with memory (B/T cells) are adaptive, not innate.
153. 153. e.g., "Passive artificial immunity — antibody injection," "Active natural immunity — natural infection."
154. 154. IgM — first antibody of the primary response.
155. 155. Complement doesn't produce antibodies — it opsonizes, lyses (MAC), and drives chemotaxis/inflammation.
156. 156. Defend against pathogens while distinguishing self from non-self.
157. 157. Immunological memory.
158. 158. Type III hypersensitivity (immune complex-mediated) — serum sickness.
159. 159. Hypothalamus (preoptic/anterior region) — thermoregulatory center.
160. 160. Vasoconstriction, shivering, ↑ metabolic heat production, piloerection — response to cold.
161. 161. Radiation — main heat loss route at rest, normal room temperature.
162. 162. Heat-loss responses (vasodilation, sweating) — anterior hypothalamus stimulation.
163. 163. Sweating and cutaneous vasodilation — prevent hyperthermia in heat.
164. 164. Voluntary behavior (seeking shade, removing clothes) — behavioral thermoregulation.
165. 165. ↑Thyroid hormone (T3/T4) — raises metabolic heat production long-term in cold.
166. 166. Local axon-reflex vasodilation/vasoconstriction (e.g., cold-induced vasodilation, "hunting reaction") — local cutaneous thermal reflex.
167. 167. High humidity reduces evaporation, so sweat drips off without cooling.
168. 168. Core temp >40°C with CNS dysfunction, often with cessation of sweating — heat stroke.

169. 169. Shivering thermogenesis + vasoconstriction (plus non-shivering thermogenesis in infants) — prevent hypothermia.
170. 170. Endogenous pyrogens (IL-1, IL-6, TNF) → hypothalamic PGE₂ → raised set point — first events of fever.
171. 171. Cerebellum (vestibulocerebellum) and vestibular nuclei — posture/equilibrium.
172. 172. Initiates/plans movement, suppresses unwanted movement — basal ganglia.
173. 173. Balance and eye-movement/postural control — vestibulocerebellum.
174. 174. Muscle spindle stretch reflex (gamma loop) — maintains muscle tone.
175. 175. Ataxia, intention tremor, dysmetria, hypotonia, nystagmus — cerebellar damage.
176. 176. Vestibulospinal tract — adjusts posture relative to gravity.
177. 177. Homeostasis — stable internal environment concept.
178. 178. e.g., insulin-glucose regulation or thermoregulation — negative feedback example.
179. 179. Nervous system — fastest control system (vs. slower endocrine system).
180. 180. Touch, pressure, temperature, pain, proprioception carried via DCML and spinothalamic tracts.
181. 181. Meissner's corpuscles (and Merkel discs) — fine touch/texture.
182. 182. Anterolateral (spinothalamic) tract — pain pathway.
183. 183. Substance P (and glutamate) — spinal pain transmission.
184. 184. Prostaglandins, bradykinin, histamine, substance P — inflammatory pain mediators.
185. 185. Sustained bilateral muscle contraction (neck/scalp), non-pulsatile pain — tension headache.
186. 186. Hypoxia ($\downarrow$ PaO₂), via carotid body chemoreceptors — main stimulus at altitude.
187. 187. Hyperventilation, respiratory alkalosis, $\uparrow$ HR, initial $\downarrow$ plasma volume — acute altitude changes.
188. 188. $\uparrow$ EPO/RBC mass (polycythemia), $\uparrow$ 2,3-BPG, $\uparrow$ capillary density, $\uparrow$ mitochondria — chronic altitude adaptation.
189. 189. Lower lean muscle mass/higher fat % — explains lower female BMR.

190. 190. $\uparrow$ HR, $\uparrow$ cardiac output, $\uparrow$ systolic BP, $\uparrow$ ventilation, blood flow shift to muscle — moderate exercise.

191. 191. Suppresses appetite/increases energy expenditure — leptin signals adequate fat stores.

192. 192. P_{aO_2} itself stays low (lower barometric pressure) even though $\uparrow$ Hb/RBC mass raises O_2 -carrying capacity.

193. 193. $\uparrow$ Red cell mass (polycythemia) — chronic altitude O_2 delivery adaptation.

194. 194. Residual volume and arterial stiffness/systolic BP — increase with aging (most other parameters decline).

195. 195. $\text{Flow} = \Delta\text{Pressure} / \text{Resistance}$ — hemodynamic "Ohm's law."

196. 196. Radius 4 (r^4) — Poiseuille's law.

197. 197. 16-fold increase — resistance $\propto 1/r^4$, so halving $r \rightarrow 2^4 = 16\times$.

198. 198. Arterial baroreceptor reflex — most important short-term BP control.

199. 199. $\uparrow$ Sympathetic outflow $\rightarrow$ tachycardia + vasoconstriction — immediate response to a BP drop.

200. 200. $\downarrow$ Hematocrit (anemia) or $\downarrow$ plasma proteins — lowers viscosity, raises flow.

201. 201. L-type Ca^{2+} channel inactivation + delayed rectifier K^+ efflux — ends the ventricular plateau.

202. 202. Extracellular Ca^{2+} influx (L-type channels) triggering CICR from the SR.

203. 203. $\uparrow$ Preload (venous return/EDV) $\rightarrow$ $\uparrow$ stroke volume — Frank-Starling relationship.

204. 204. Funny current (I_f , HCN channels) (plus T-type Ca^{2+}) — Phase 4 pacemaker depolarization.

205. 205. Ventricles relax, semilunar valves close, pressure falls rapidly before AV valves open — volume constant — isovolumetric relaxation.

206. 206. Cardiac muscle requires extracellular Ca^{2+} entry (CICR) and is a functional syncytium (gap junctions); skeletal doesn't need extracellular Ca^{2+} .

207. 207. Long absolute refractory period (from the prolonged plateau) prevents summation/tetany.

208. 208. Sympathetic (β_1) stimulation and vagal withdrawal — increase heart rate.

209. 209. Venous return (preload), via the Frank-Starling mechanism — key immediate determinant of CO/SV.

210. 210. $\uparrow$ Afterload or extreme $\uparrow$ HR ($\downarrow$ filling time) — can decrease CO.

211. 211. Intrinsic length-tension property: $\uparrow$ stretch (preload) $\rightarrow$ $\uparrow$ contractile force, matching CO to venous return.
212. 212. $\uparrow$ BP $\rightarrow$ $\uparrow$ baroreceptor firing $\rightarrow$ $\downarrow$ sympathetic/ $\uparrow$ parasympathetic $\rightarrow$ $\downarrow$ HR/BP — fast, negative feedback, resets with chronic hypertension.
213. 213. CO regulated by preload, afterload, contractility, and HR, matching venous return.
214. 214. $\uparrow$ HR and peripheral vasoconstriction — baroreceptor response on standing.
215. 215. $\uparrow$ Sympathetic vasoconstriction ($\uparrow$ TPR) — most effective short-term MAP raiser.
216. 216. $\uparrow$ Venous return (Frank-Starling) and circulating catecholamines — CO can still rise in a denervated heart.
217. 217. Arterioles — resistance vessels.
218. 218. Veins — capacitance vessels.
219. 219. Vessel radius (r) — greatest influence on flow.
220. 220. Capillaries — highest total cross-sectional area, lowest velocity.
221. 221. Capillaries — small radius keeps wall tension low (Laplace's law) despite thin walls.
222. 222. Arterioles — biggest contributor to total peripheral resistance.
223. 223. Diaphragm (with external intercostals) — quiet inspiration.
224. 224. Passive elastic recoil of lungs/chest wall — normal quiet expiration.
225. 225. Intrapleural pressure — always negative in quiet breathing.
226. 226. Arterial PCO_2 (via central chemoreceptors/CSF H^+) — main respiratory drive normally.
227. 227. Reduces alveolar surface tension; produced by type II pneumocytes; stabilizes alveoli — surfactant.
228. 228. Inversely proportional — $\uparrow$ alveolar ventilation $\rightarrow$ $\downarrow$ $PaCO_2$.
229. 229. Thoracic expansion (diaphragm contraction) increasing lung volume/elastic recoil — more negative intrapleural pressure.
230. 230. Perfusion falls more than ventilation from base to apex (gravity) — higher V/Q at apex.
231. 231. $\uparrow$ CO_2/H^+ in tissue capillaries lowers Hb- O_2 affinity, promoting O_2 unloading (Bohr effect).

232. 232. Residual volume (and FRC, TLC, which depend on it) — not measurable by simple spirometry.
233. 233. Hypothermia/ ↓ 2,3-BPG/alkalosis left-shift the curve — NOT rightward shifters.
234. 234. ~75% — Hb saturation at PO₂ ~40 mmHg (mixed venous).
235. 235. ~98% of O₂ is bound to hemoglobin, small fraction dissolved in plasma.
236. 236. Humidification and continuous gas exchange (O₂ uptake/CO₂ addition) dilute alveolar PO₂ below inspired PO₂.
237. 237. V/Q mismatch, diffusion impairment, right-to-left shunt — increase the A-a gradient.
238. 238. ~250 mL/min — resting VO₂.
239. 239. Bicarbonate (~70%) — main CO₂ transport form.
240. 240. Ratio of O₂ consumption to O₂ delivery — determines the O₂ extraction ratio.
241. 241. Medulla oblongata (pre-Bötzinger complex/dorsal respiratory group) — respiratory rhythm generator.
242. 242. CO₂ (via H⁺ in CSF) — main central chemoreceptor stimulus.
243. 243. Opioids (also barbiturates/sedatives) — depress respiratory centers, cause arrest in overdose.
244. 244. Lung stretch receptors inhibit further inspiration when overinflated — Hering-Breuer reflex.
245. 245. Peripheral (carotid body) chemoreceptor response to hypoxemia — initial altitude ventilation increase.
246. 246. Rising PaCO₂/falling pH (central chemoreceptor drive) — main trigger to resume breathing after breath-holding.
247. 247. Renal excretion of HCO₃⁻ normalizes CSF pH, removing the brake so hypoxic drive keeps ventilation elevated.
248. 248. Promotes sustained inspiration — apneustic center (lower pons).
249. 249. Lung inflation → stretch receptors → vagal inhibition of inspiration — Hering-Breuer inflation reflex.
250. 250. Steroid and thyroid hormones — act via intracellular receptors on gene transcription.
251. 251. IP₃/DAG-Ca²⁺ (phospholipase C pathway) — second messenger for V1/α1-receptor signaling.
252. 252. Growth hormone (and prolactin) — use JAK-STAT signaling.

253. 253. Oxytocin and ADH — made in hypothalamus, released from posterior pituitary.
254. 254. TRH → TSH → T3/T4, with T3/T4 negative feedback on TRH and TSH.
255. 255. ↓ Serum calcium by inhibiting osteoclasts (minor role in adult humans) — calcitonin, from thyroid C- cells.
256. 256. Lipophilic hormone diffuses in, binds intracellular receptor, complex acts as a transcription factor.
257. 257. Lactotrophs (prolactin) — under predominantly inhibitory (dopamine) hypothalamic control.
258. 258. TSH, LH, FSH, hCG — share a common α -subunit, unique β -subunit gives specificity.
259. 259. Central (neurogenic) diabetes insipidus.
260. 260. ↑ Ca^{2+} reabsorption (distal tubule), ↓ phosphate reabsorption, activates vitamin D — PTH renal actions.
261. 261. Primary hyperparathyroidism.
262. 262. ACTH (from CRH-driven anterior pituitary) — stimulates cortisol.
263. 263. ↑ Gluconeogenesis/blood glucose, anti-inflammatory, protein catabolism — cortisol effects.
264. 264. Angiotensin II (and plasma K^+) — major aldosterone regulators.
265. 265. Glomerulosa–mineralocorticoids, fasciculata– glucocorticoids, reticularis–androgens.
266. 266. High cortisol with LOW (suppressed) ACTH — adrenal-tumor Cushing's.
267. 267. PNMT (phenylethanolamine N- methyltransferase) — NE → Epi, cortisol-induced.
268. 268. β -adrenergic receptors (β_2 liver glycogenolysis; β_1/β_3 adipocyte lipolysis).
269. 269. Sympathetic preganglionic (cholinergic, nicotinic) stimulation — triggers catecholamine release in stress.
270. 270. Hypertension, hypokalemia, metabolic alkalosis, low renin — primary hyperaldosteronism.
271. 271. Episodic massive catecholamine release from the tumor — pheochromocytoma's paroxysmal symptoms.
272. 272. Cholesterol → pregnenolone (cholesterol side- chain cleavage/desmase, StAR-mediated) — rate-limiting step.
273. 273. 11β -hydroxylase deficiency — DOC accumulation, hypertension, low cortisol.

274. 274. Ovulation (completion of meiosis I, follicle rupture) — triggered by LH surge.
275. 275. Progesterone (from corpus luteum) — maintains secretory endometrium.
276. 276. Day 14 — LH surge in a 28-day cycle.
277. 277. Oogonia arrest in prophase I before birth; meiosis I completes just before ovulation; meiosis II completes only at fertilization.
278. 278. Corpus luteum — main progesterone source, ■rst trimester.
279. 279. Human placental lactogen (hPL).
280. 280. Metaphase II — secondary oocyte arrest.
281. 281. Sustained high estradiol switches hypothalamic- pituitary feedback from negative to positive, triggering the LH surge.
282. 282. Falling progesterone (and estrogen) — ■rst hormonal change if corpus luteum fails.
283. 283. Progesterone/estrogen withdrawal → spiral artery constriction → endometrial ischemia/shedding.
284. 284. Rising estrogen initially gives negative feedback, then switches to positive feedback near ovulation.
285. 285. Human placental lactogen (hPL).
286. 286. hCG secretion by trophoblast — earliest endocrine change, maintains corpus luteum.
287. 287. Placenta relies on fetal/maternal precursors (DHEA-S) for estrogen synthesis — the "fetoplacental unit."
288. 288. Progesterone (with estrogen, prolactin, hPL) — mammary lobuloalveolar growth.
289. 289. Spermatogonia — germ cells that undergo spermatogenesis.
290. 290. Testosterone (with FSH) — critical for initiating/maintaining spermatogenesis.
291. 291. Sertoli cells — FSH target supporting spermatogenesis.
292. 292. LH — stimulates Leydig cells to make testosterone.
293. 293. Peripheral tissues (prostate, genital skin), via 5 α - reductase — testosterone → DHT.
294. 294. Inhibin (from Sertoli cells) — inhibits FSH.
295. 295. Spermatogonia → primary spermatocyte → secondary spermatocyte → spermatid → spermatozoon.

296. 296. Sertoli-cell tight junctions isolate germ cells from the immune system — blood-testis barrier.
297. 297. Testosterone inhibits GnRH/LH; inhibin inhibits FSH — negative feedback loop.
298. 298. Stimulates gastric HCl/pepsinogen and gastrin release — cephalic phase vagal effect (same as Q101).
299. 299. Loss of vagal receptive relaxation — impaired liquid emptying post-vagotomy (same as Q102).
300. 300. Hypotonic, low-volume, mucous-rich resting saliva (same as Q103).
301. 301. Peristaltic reflex — contraction proximal, relaxation distal to a bolus (same as Q104).
302. 302. Non-specific, immediate, present from birth — innate immunity (same as Q147).
303. 303. Antigen specificity and memory — adaptive vs innate (same as Q148).
304. 304. T lymphocytes — cell-mediated immunity (same as Q149).
305. 305. Secondary response is faster, larger, and IgG- predominant (memory B cells) vs slower IgM- predominant primary response.
306. 306. Antigen-specific lymphocytes with memory are adaptive, not innate (same as Q152).
307. 307. e.g., "Passive artificial immunity — antibody injection" (same as Q153).
308. 308. IgM — first antibody, primary response (same as Q154).
309. 309. Complement doesn't make antibodies (same as Q155).
310. 310. Passive natural immunity — maternal IgG crossing the placenta.
311. 311. Cell-mediated (T-cell) immunodeficiency — e.g., DiGeorge syndrome, thymic aplasia.
312. 312. APCs process/present antigen via MHC to T cells, initiating adaptive immunity.
313. 313. Type III hypersensitivity (immune complexes) — serum sickness (same as Q158).
314. 314. Hypothalamus — thermoregulatory center (same as Q159).
315. 315. Vasoconstriction, shivering, ↑ heat production — cold response (same as Q160).
316. 316. Radiation — resting heat loss (same as Q161).

317. 317. Heat-loss responses (vasodilation, sweating) — anterior hypothalamus (same as Q162).
318. 318. Sweating and cutaneous vasodilation — heat response (same as Q163).
319. 319. Voluntary behavior — behavioral thermoregulation (same as Q164).
320. 320. ↑Thyroid hormone — long-term cold response (same as Q165).
321. 321. Local axon-reflex vasomotor response — cutaneous thermal reflex (same as Q166).
322. 322. High humidity blocks evaporative cooling (same as Q167).
323. 323. Core temp >40°C with CNS dysfunction — heat stroke (same as Q168).
324. 324. Shivering + vasoconstriction — prevents hypothermia (same as Q169).
325. 325. Endogenous pyrogens → hypothalamic PGE2 → raised set point — fever onset (same as Q170).
326. 326. Cerebellum/vestibular nuclei — posture/equilibrium (same as Q171).
327. 327. Initiates/refines movement — basal ganglia (same as Q172).
328. 328. Balance/eye-movement control — vestibulocerebellum (same as Q173).
329. 329. Muscle spindle gamma loop — muscle tone (same as Q174).
330. 330. Ataxia, intention tremor, dysmetria, hypotonia — cerebellar damage (same as Q175).
331. 331. Vestibulospinal tract — balance/posture (same as Q176).
332. 332. Homeostasis (same as Q177).
333. 333. e.g., insulin-glucose feedback (same as Q178).
334. 334. Nervous system — fast control (same as Q179).
335. 335. Touch, pressure, temperature, pain, proprioception — somatic sensations (same as Q180).
336. 336. Meissner's corpuscles — fine touch/texture (same as Q181).
337. 337. Spinothalamic (anterolateral) tract — pain (same as Q182).
338. 338. Substance P — spinal pain neurotransmitter (same as Q183).
339. 339. Prostaglandins/bradykinin/histamine — inflammatory pain mediators (same as Q184).

340. 340. Sustained muscle contraction, bilateral, non- pulsatile — tension headache (same as Q185).
341. 341. Hypoxia via carotid bodies — altitude ventilation stimulus (same as Q186).
342. 342. Hyperventilation, respiratory alkalosis, $\uparrow$ HR — acute altitude change (same as Q187).
343. 343. $\uparrow$ RBC mass, $\uparrow$ 2,3-BPG, $\uparrow$ capillary density — chronic altitude adaptation (same as Q188).
344. 344. Lower lean muscle mass — lower female BMR (same as Q189).
345. 345. $\uparrow$ HR, CO, systolic BP, ventilation — moderate exercise (same as Q190).
346. 346. Suppresses appetite — leptin (same as Q191).
347. 347. PaO_2 itself remains low despite $\uparrow$ ventilation — altitude $\uparrow$ PaO_2 (same as Q192).
348. 348. $\uparrow$ Red cell mass — O_2 delivery adaptation (same as Q193).
349. 349. Residual volume/arterial stiffness increase with age (same as Q194).
350. 350. $\text{Flow} = \Delta\text{Pressure}/\text{Resistance}$ (same as Q195).